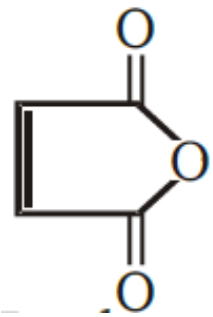
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1



$\xrightarrow{H_3O^{\oplus}} (A) \xrightarrow{NaOH} (B) \xrightarrow{\text{Kolbe electrolysis}} (C)$

Product (C) of above reaction is:

(A) $H_2C = CH_2$

(B) $CH_3 - C \equiv C - CH_3$

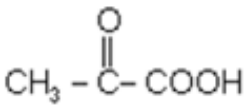
(C) $HC \equiv CH$

(D) $CH_3 - CH = CH - CH_3$

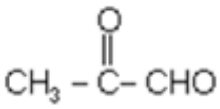
Q.2

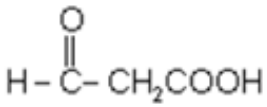
End product of the following sequence of reactions is

$$CH \equiv CH \xrightarrow{CH_3MgBr} \xrightarrow{CO_2 / H_3O^+} \xrightarrow{HgSO_4 / H_2SO_4} \xrightarrow[\Delta]{Ag_2O}$$

(A) 

(B) $CH_2(COOH)_2$

(C) 

(D) 

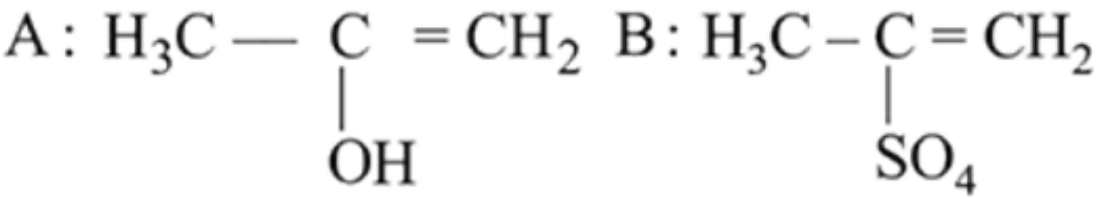
Q.3

Predict the correct intermediate and product in the following reaction:

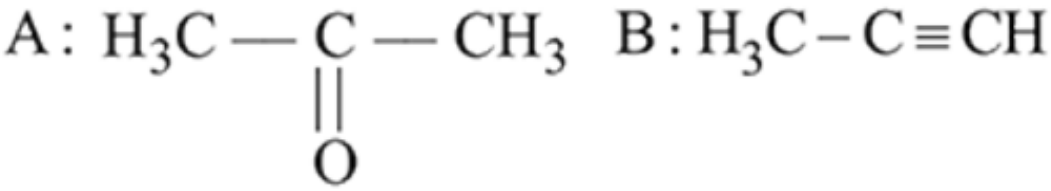
$$H_3C - C \equiv CH \xrightarrow[HgSO_4]{H_2O,\ H_2SO_2} \underset{(A)}{\text{Intermediate}} \longrightarrow \underset{(B)}{\text{product}}$$

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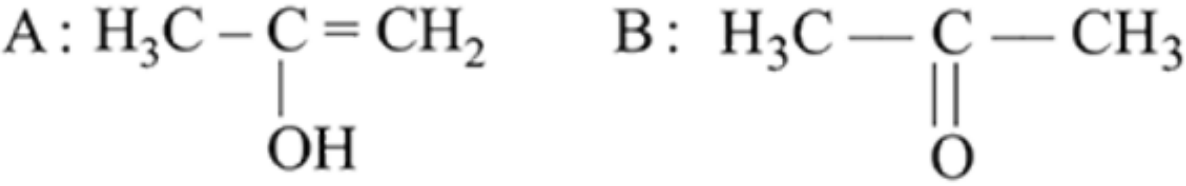
(A)



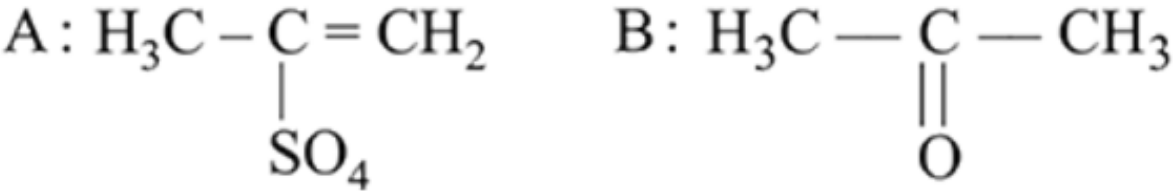
(B)



(C)



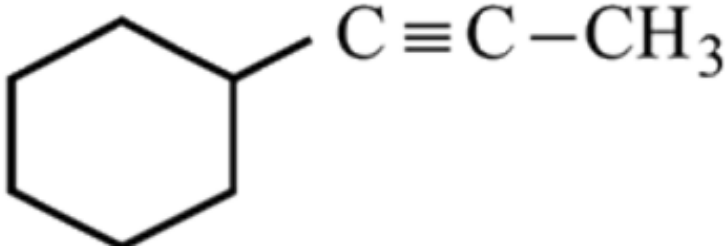
(D)



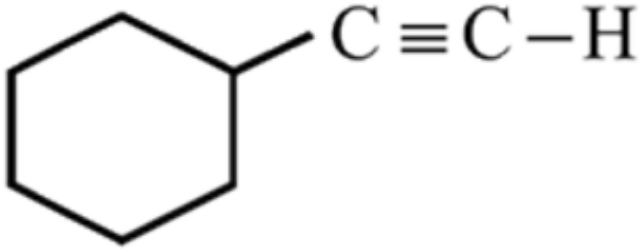
Q.4

Which of the following will form alkynide?

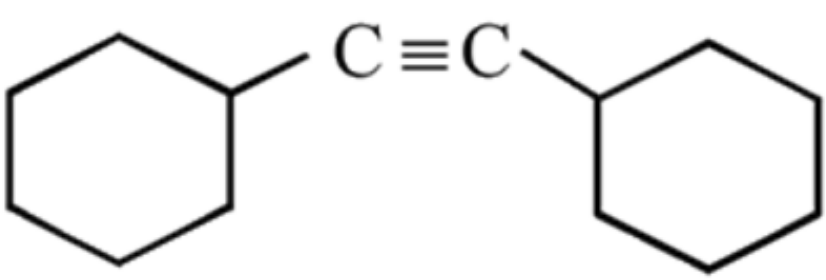
(A)



(B)



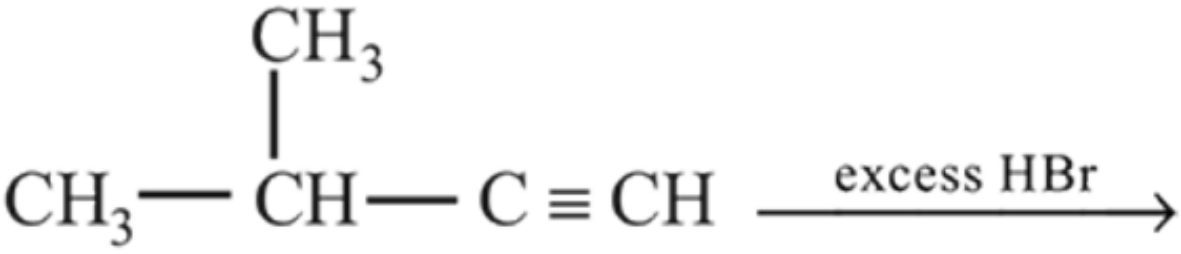
(C)



(D) $CH_3 - C \equiv C - CH_3$

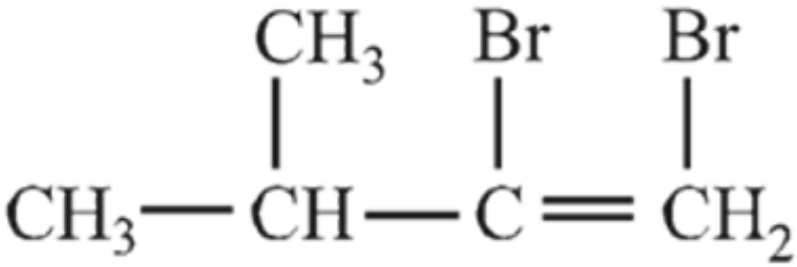
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Q.5

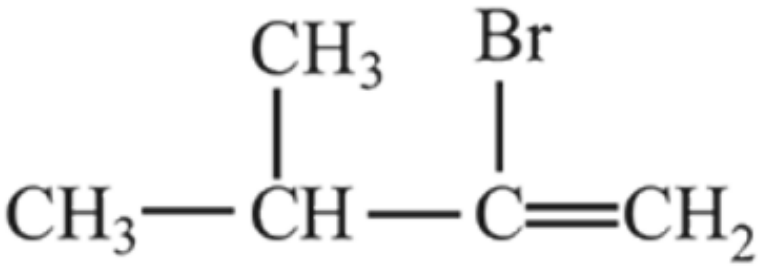


The product of the above reaction is:

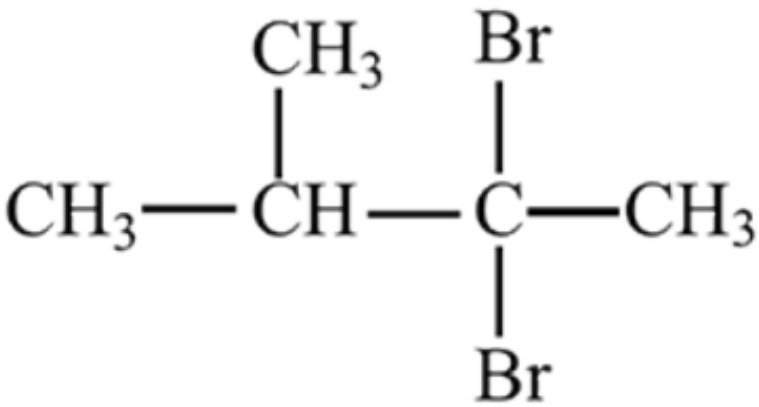
(A)



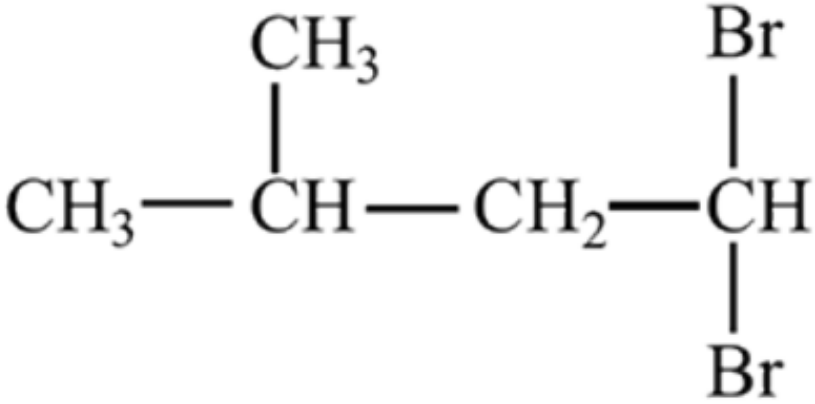
(B)



(C)



(D)



Q.6

What is the end product of the following

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sequences of operations?

$$CaC_2 \xrightarrow{H_2O} A \xrightarrow[Hg^{2+}]{dil.H_2SO_4} B \xrightarrow[H_2]{Ni} C$$

(A) Methyl alcohol

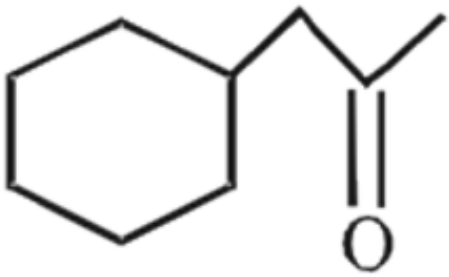
(B) Acetaldehyde

(C) C_2H_5OH

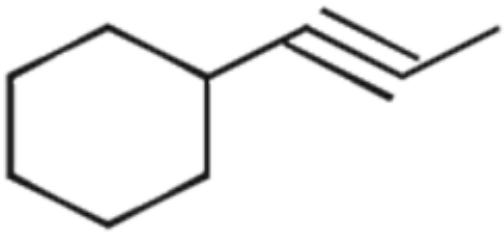
(D) C_2H_4

Q.7

Choose the correct alkyne and reagents for the preparation of

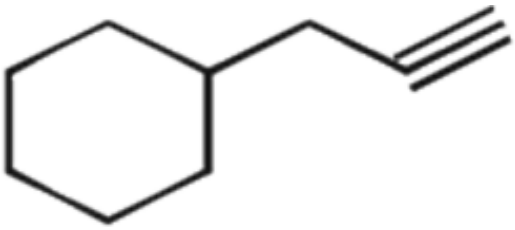


(A)



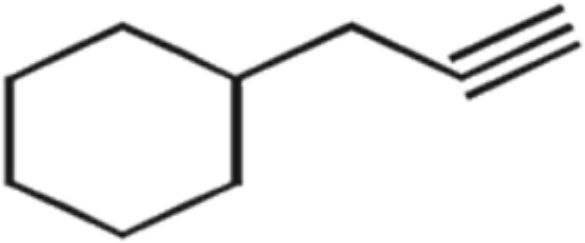
$, HgSO_4, \ H_2SO_4, \ H_2O$

(B)



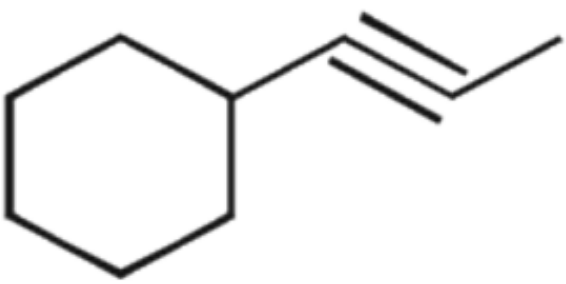
$, HgSO_4, \ H_2SO_4, \ H_2O$

(C)



$, BH_3, \ H_2O_2, NaOH$

(D)

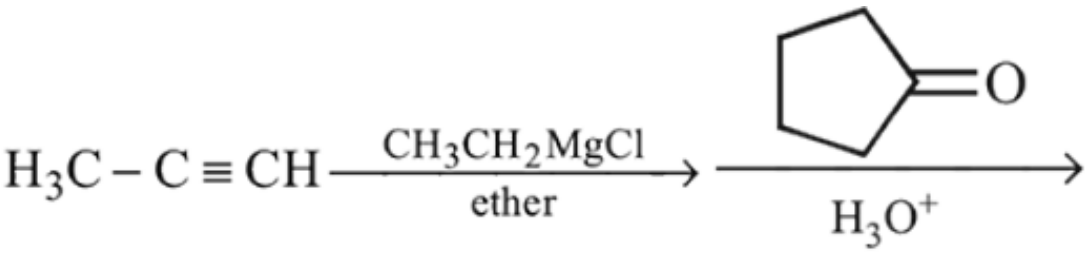


$, BH_3, \ H_2O_2, NaOH$

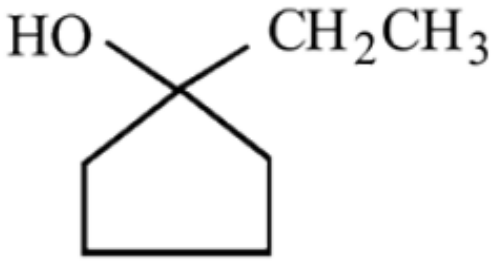
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Q.8

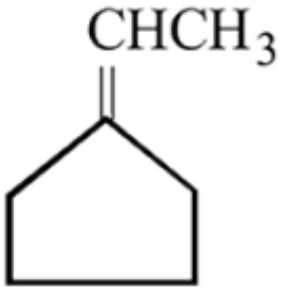
The major product of the following reaction



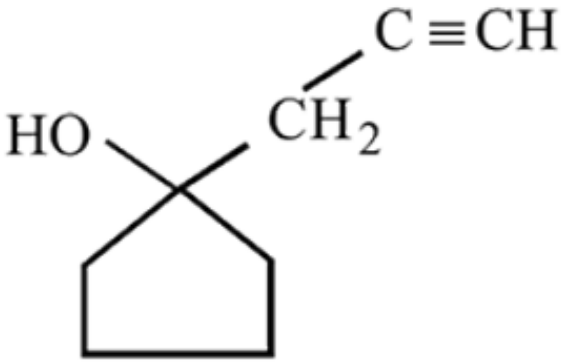
(A)



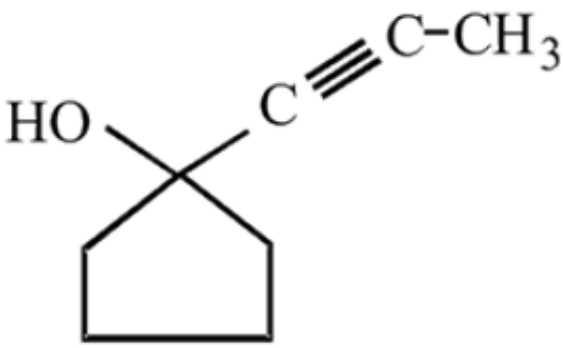
(B)



(C)



(D)



Q.9

A compound (X) (C_5H_8) reacts with ammonical $AgNO_3$ to give a white precipitate, and on oxidation with hot alkaline $KMnO_4$ gives the acid, $(CH_3)_2\,CHCOOH$, therefore X is –

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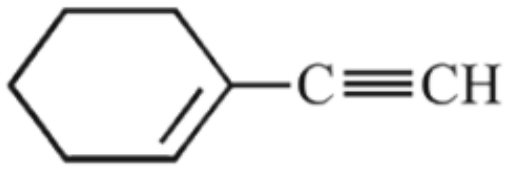
(A) $CH_2 = CH - CH = CH - CH_3$

(B) $CH_3 - CH = CH - CH_2 - CH_3$

(C) $(CH_3)_2\,CH - C \equiv CH$

(D) $(CH_3)_2\,C = C = CH_2$

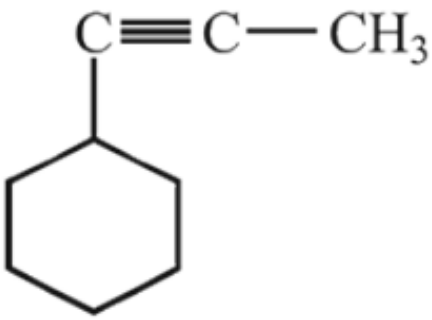
Q.10



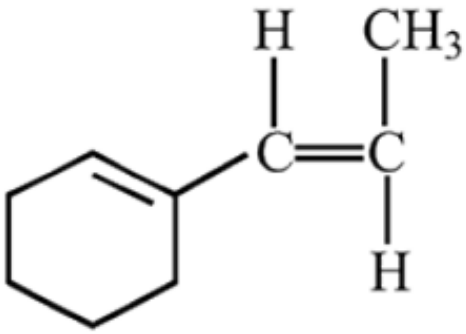
$$\xrightarrow[\text{(ii) } CH_3Br]{\text{(i) } NaNH_2,\ NH_3} (A) \xrightarrow[\text{Lindlar catalyst}]{H_2} (B)\ ;$$

Product (B) is:

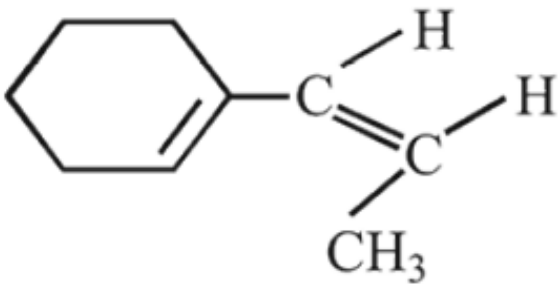
(A)



(B)

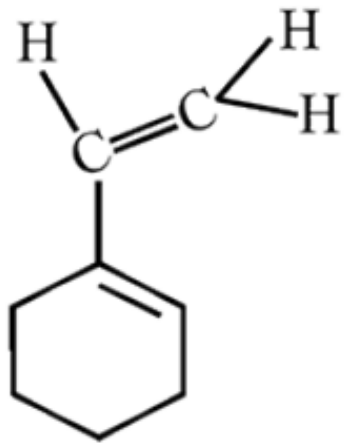


(C)



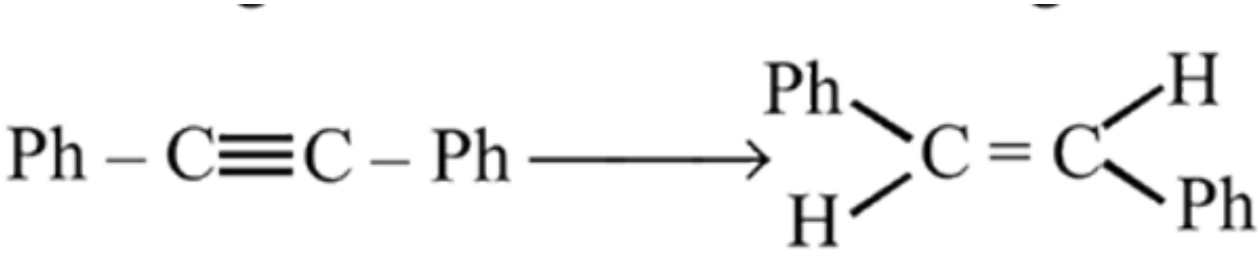
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(D)



Q.11

The reagent needed for converting



(A) Cat. Hydrogenation

(B) H_2/ Lindlar Cat.

(C) Li/NH_3

(D) $LiAlH_4$

Q.12

An unknown compound A has a molecular formula C_4H_6, when A is treated with an excess of Br_2, a new substance B with formula $C_4H_6Br_2$ is formed. A forms a white precipitate with ammonical silver nitrate solution. A may be

(A) Butyne-1

(B) Butyne-2

(C) Butene-1

(D) Butene-2

Q.13

Which one of the following class of compounds is obtained by polymerization of acetylene?

(A) Poly-yne

(B) Poly-ene

(C) Poly-ester

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(D) Poly-amine

Q.14

The hydration of propyne results in formation of

(A) Acetone

(B) Propanol-1

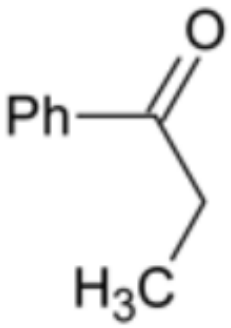
(C) Propene

(D) Propanal

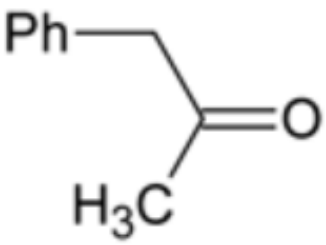
Q.15

In the reacton $\mathrm{Ph-C\equiv C-CH_3} \xrightarrow{\mathrm{Hg^{2+}H^+}} \mathrm{A}$
The product A is:

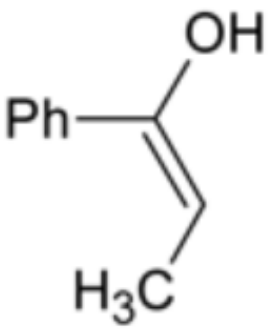
(A)



(B)

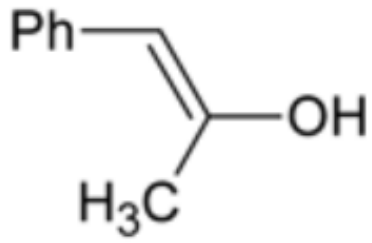


(C)



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(D)



Q.16

2-Buiyne which treated with litium in presence of liquld ammonia glye8

(A) cls-2-butene

(B) trens-2butent

(C) nbutane

(D) 1-butyne

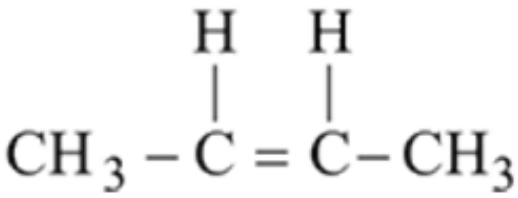
Q.17

The treatment of CH_3MgX with $CH_3C \equiv C - H$ produces

(A) $CH_3 - CH = CH_2$

(B) $CH_3C \equiv C - CH_3$

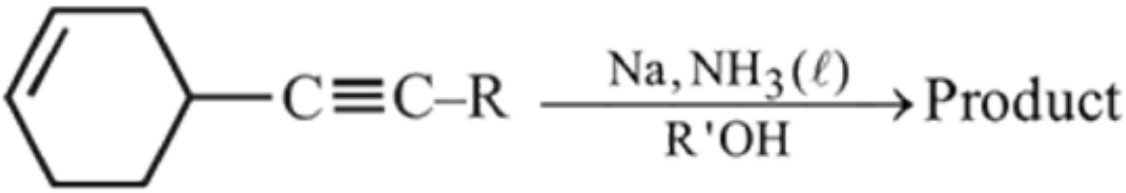
(C)



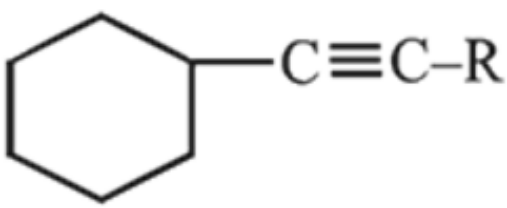
(D) CH_4

Q.18

The main product of the reaction is :

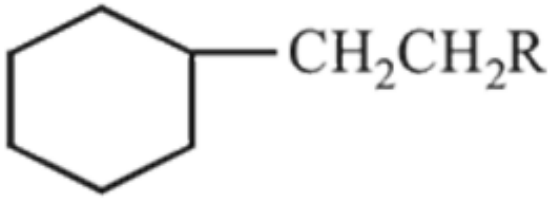


(A)

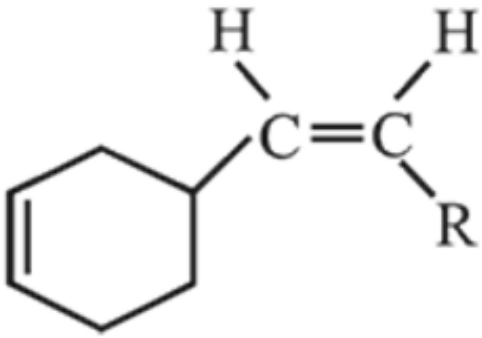


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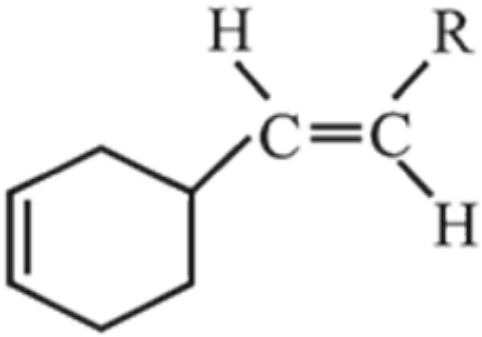
(B)



(C)



(D)



Q.19

Identify the reagent from the following list which can easily distinguish between 1 -butyne and 2-butyne

(A) bromine, CCl_4

(B) H_2, Lindlar catalyst

(C) dilute $H_2SO_4, HgSO_4$

(D) ammonical Cu_2Cl_2 solution

Q.20

$$Ph-\underset{\displaystyle Cl}{\overset{\displaystyle Cl}{\overset{|}{\underset{|}{C}}}}-CH_3 \xrightarrow{3NaNH_2} (A);$$ Product (A) is : Product

(A) $Ph - CH = CH_2$

(B) $Ph - C \equiv CH$

(C) $Ph - CH_2 - CH_3$

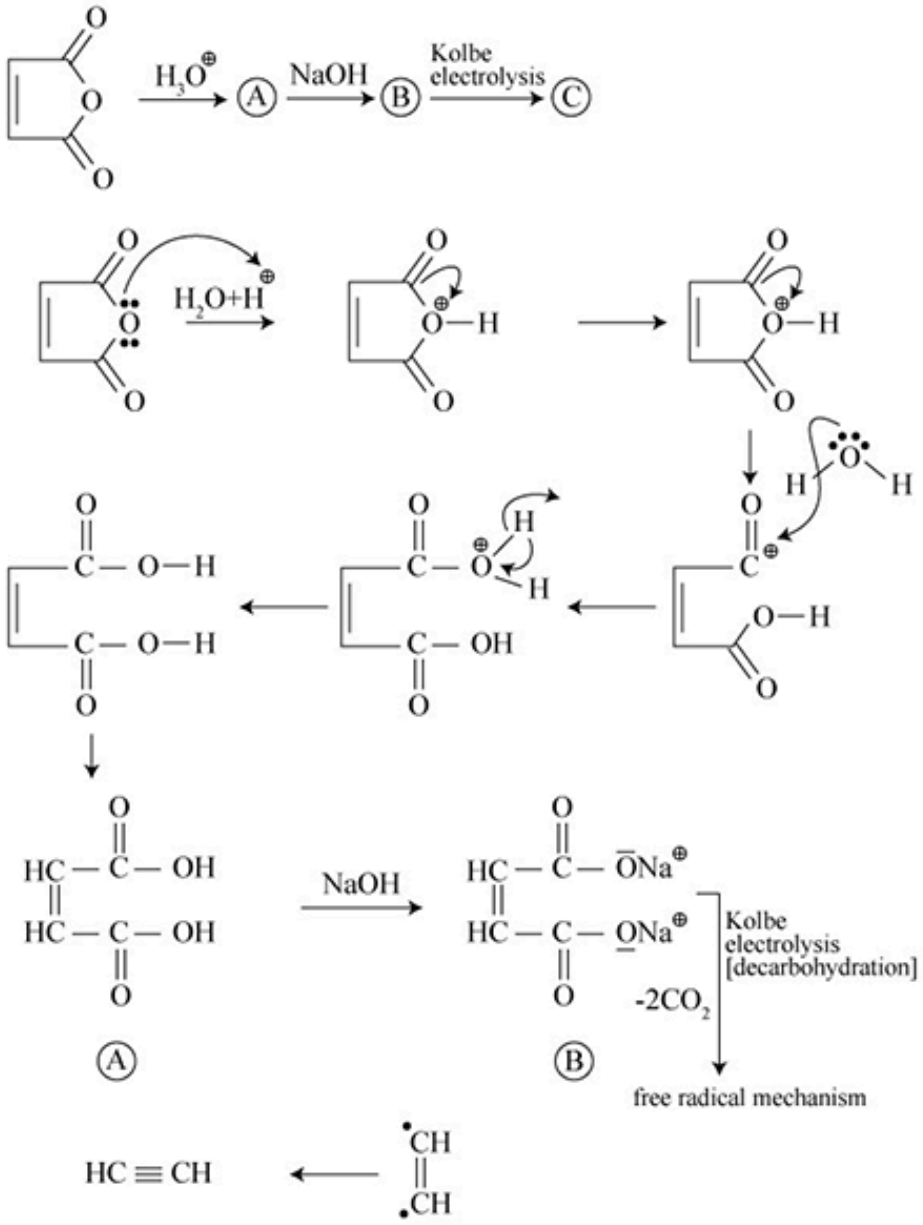
(D) $Ph - C \equiv \overset{\ominus\,\oplus}{C}N$ a

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Q.1 Answer:

$HC \equiv CH$

Solution:

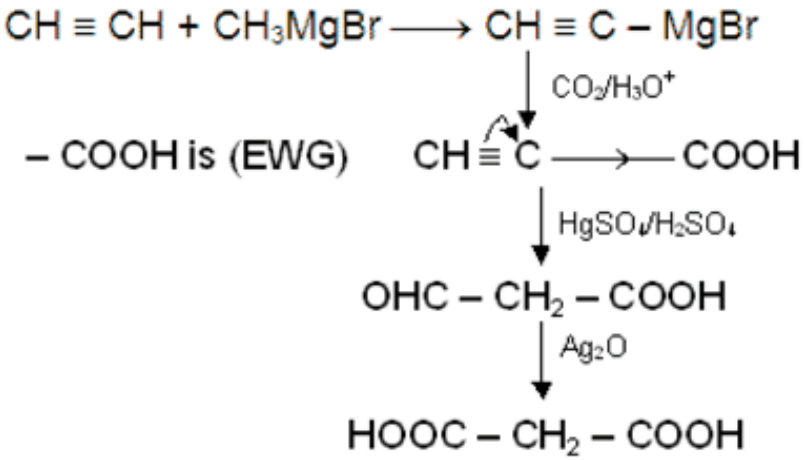


Q.2 Answer:

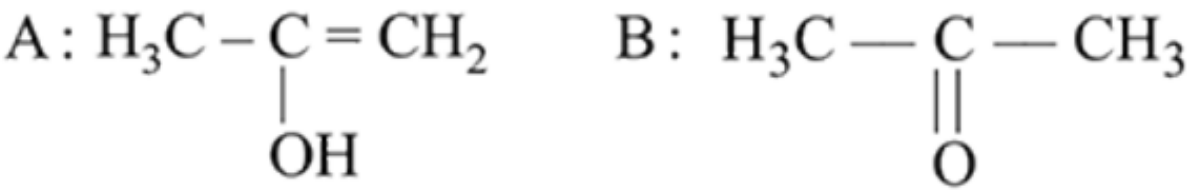
$CH_2(COOH)_2$

Solution:

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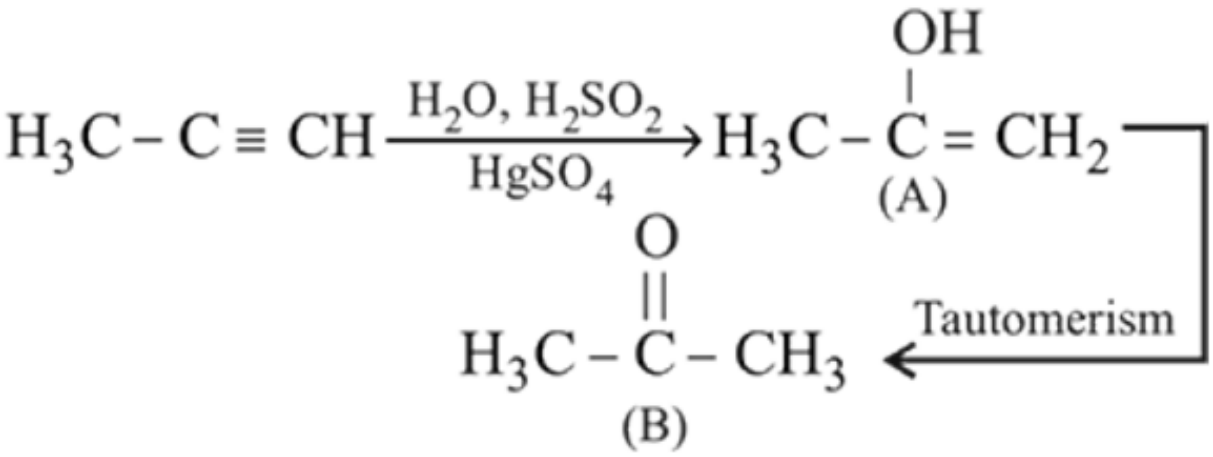


Q.3 Answer:

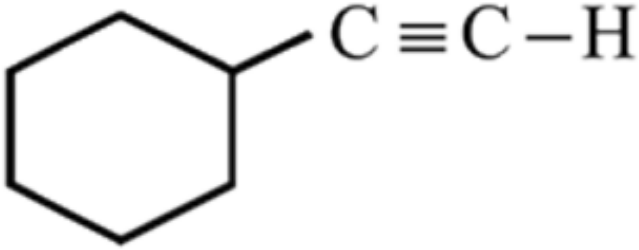


Solution:

Hydration of alkynes give ketones.

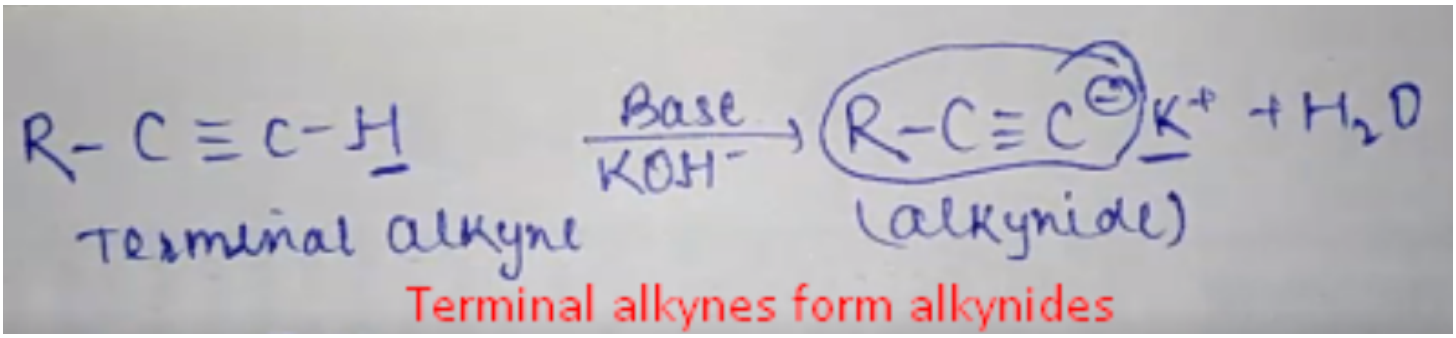


Q.4 Answer:

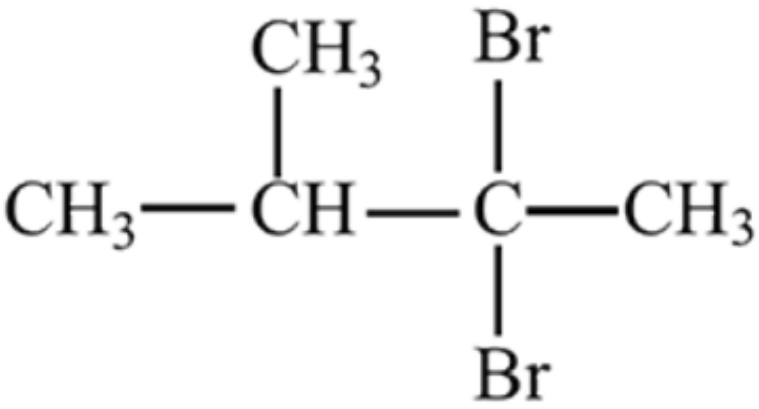


Solution:

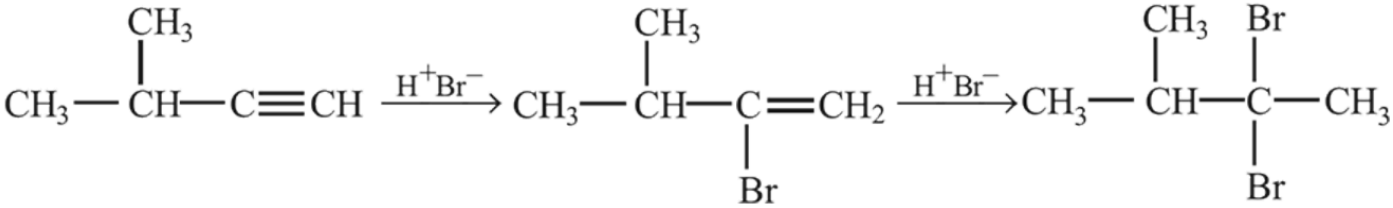
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Q.5 Answer:



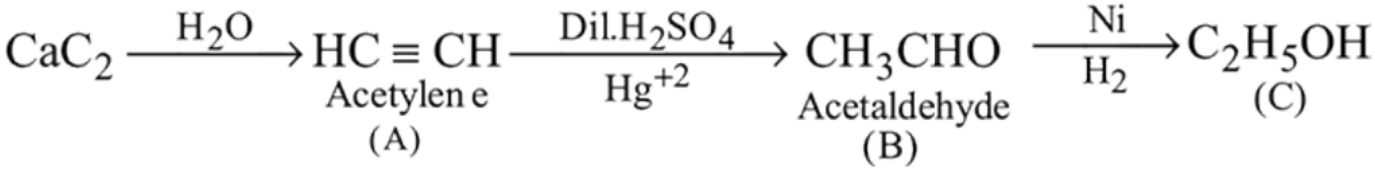
Solution:



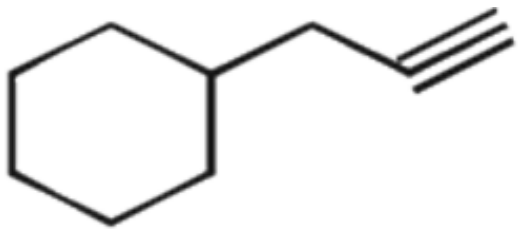
Q.6 Answer:

C_2H_5OH

Solution:



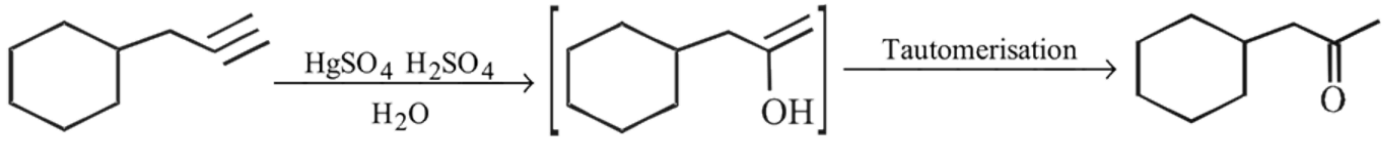
Q.7 Answer:



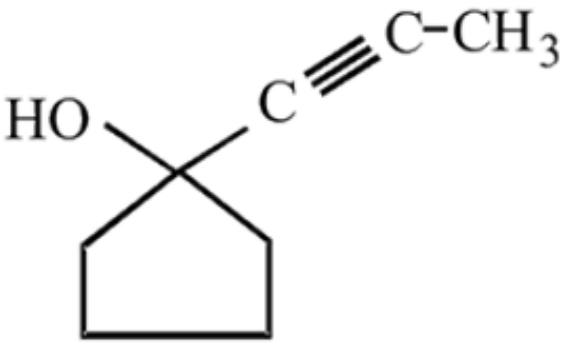
, $HgSO_4$, H_2SO_4, H_2O

Solution:

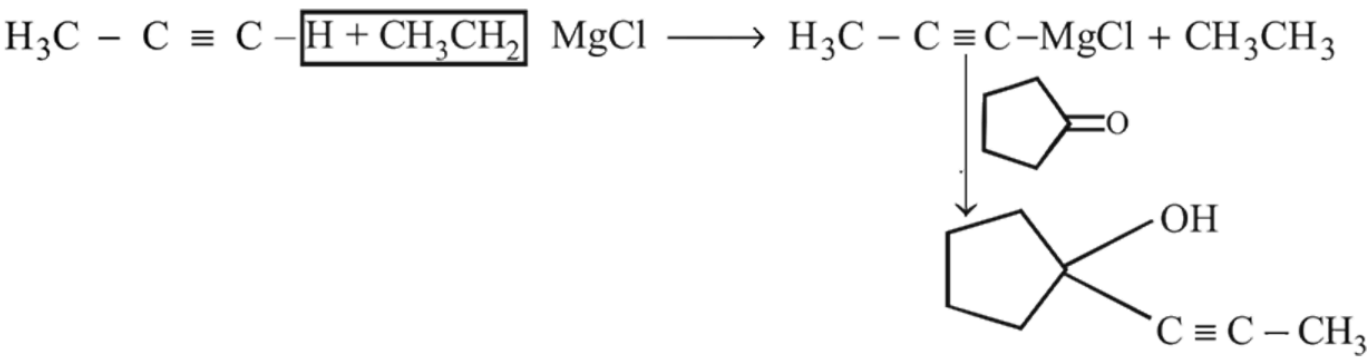
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Q.8 Answer:



Solution:



Q.9 Answer:

$(CH_3)_2\,CH - C \equiv CH$

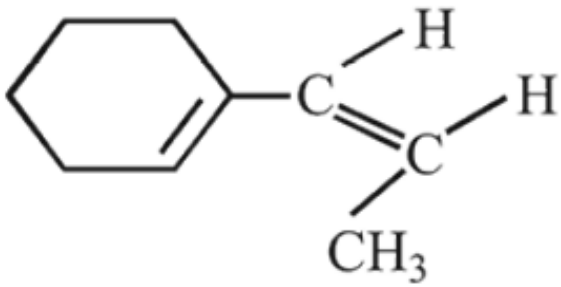
Solution:

Ammonical silver nitrate (Tollens reagent) reacts with 1-alkyne only and give white ppt so reactant should be terminal alkyne.

Reaction with hot $KMnO_4$ is:

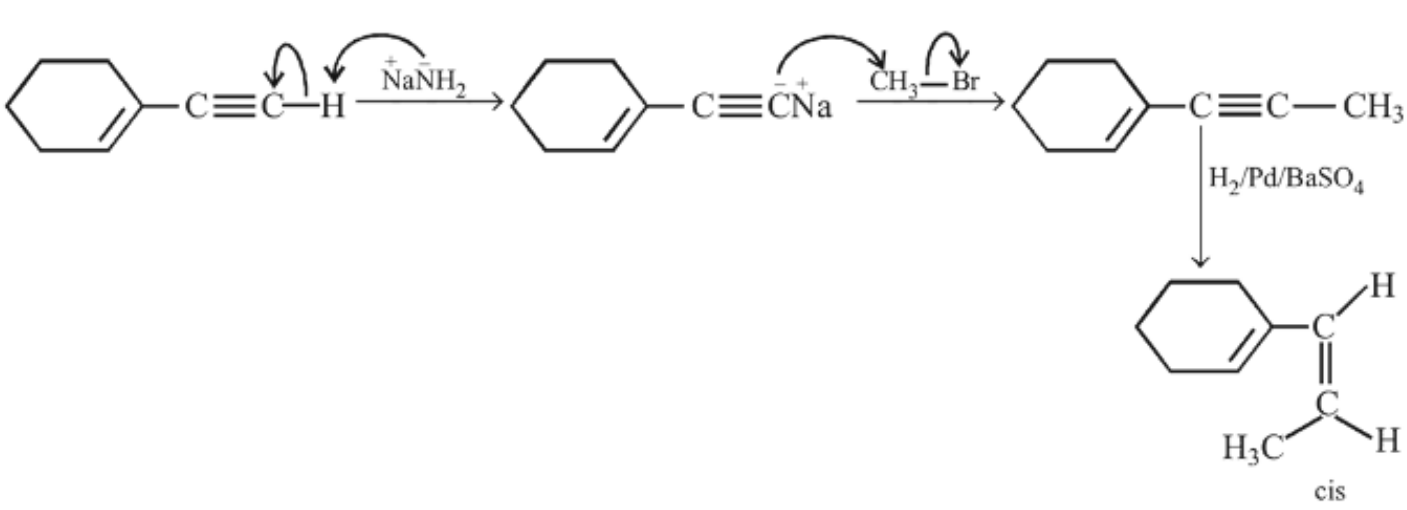
$$(CH_3)_2\,CH - C \equiv CH \rightarrow (CH_3)_2\,CHCOOH + CO_2$$

Q.10 Answer:



Solution:

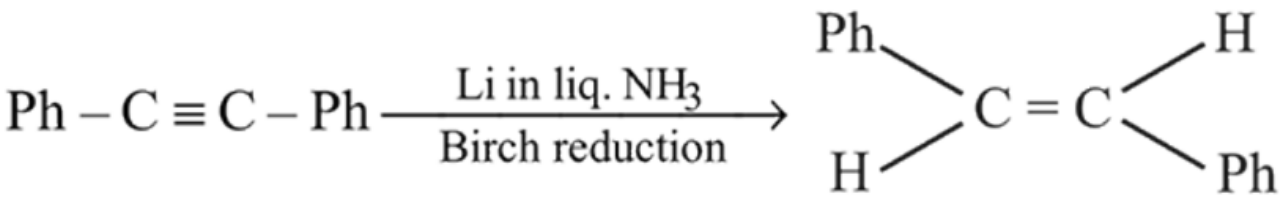
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Q.11 Answer:

Li/NH_3

Solution:

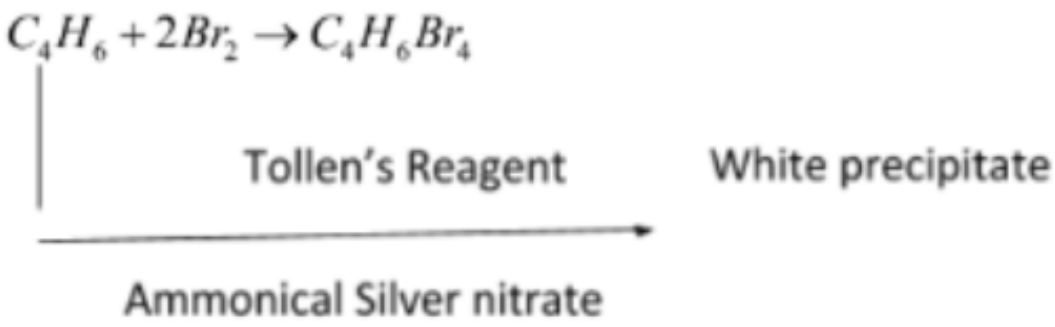


Q.12 Answer:

Butyne-1

Solution:

Let us form an equation according to the given condition



C_4H_6 **is butyne. It can either be But-1-yne or But-2-yne.**

$CH_3 - CH_2 - C \equiv CH$ **But-1-yne**

$CH_3 - C \equiv C - CH_3$ **But-2-yne**

If the triple bond is present with the carbon atom 1 and 2 , then the hydrogen atom is present at C-atom 1 so it is acidic in nature.

$CH_3 - CH_2 - C \equiv CH$ **is acidic in nature.**

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Since Ammoniacal silver nitrate is reducing agent, it reduces butyne, and butyne oxidize ammonical silver, nitrate solution.

We know that oxidation and Reduction occur together.

But-1-yne reduces by tollen's reagent and form white precipitate

Q.13 Answer:

Poly-ene

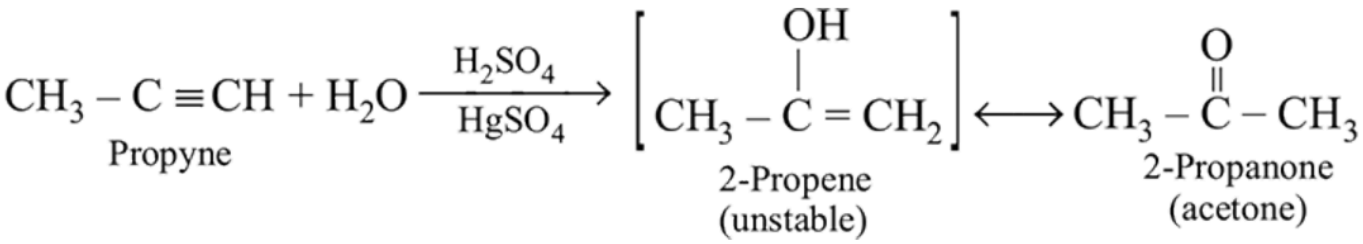
Solution:

![rendered image](https://i.jee-adv.eu.org/images/45e62b092dd29907d8aeabe816c8ad7c774fd5da6b32187cb049e1b993e93932.png)

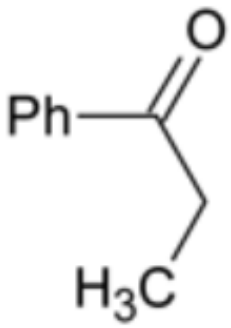
Q.14 Answer:

Acetone

Solution:

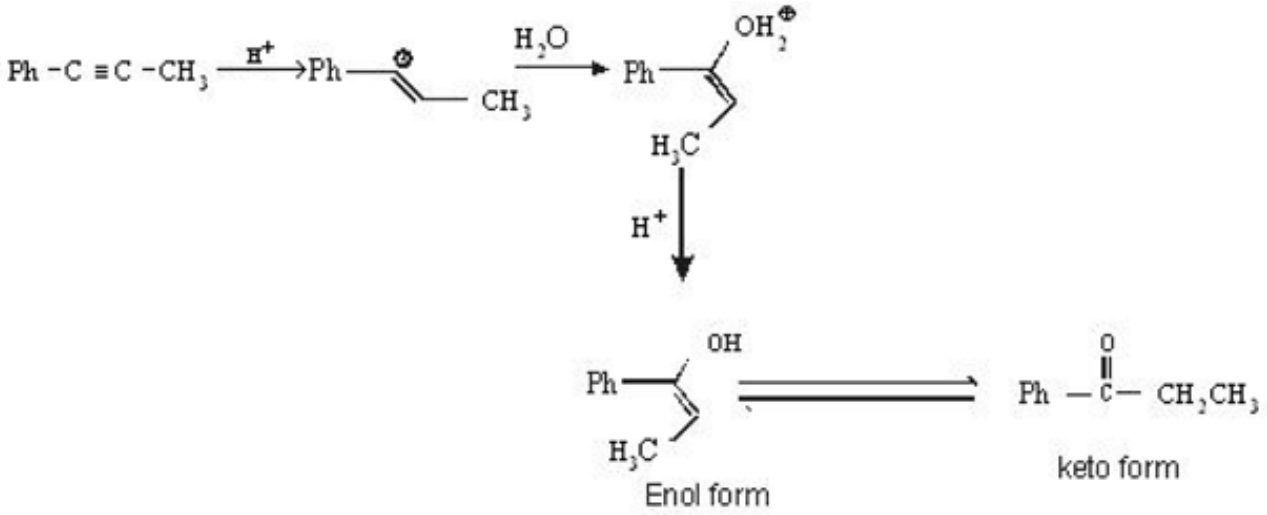


Q.15 Answer:



Solution:

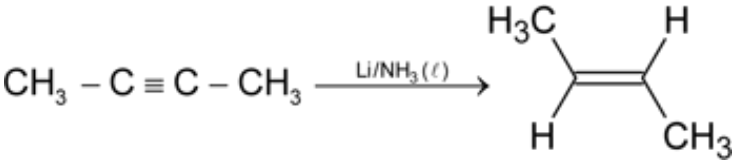
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Q.16 Answer:

trens-2butent

Solution:



Q.17 Answer:

CH_4

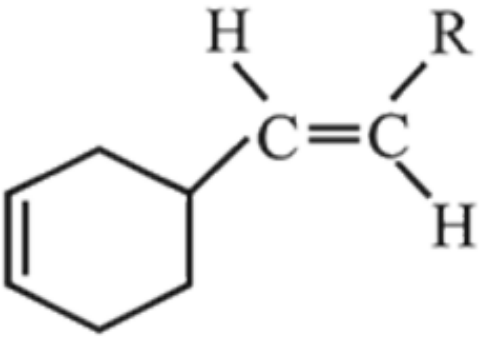
Solution:

Writing the reaction we get

$$CH_3MgX + CH_3 - C \equiv C - H \longrightarrow CH_3 - C \equiv CMgX + CH_4(g)$$

So we find that CH_4 is produced in this reaction.

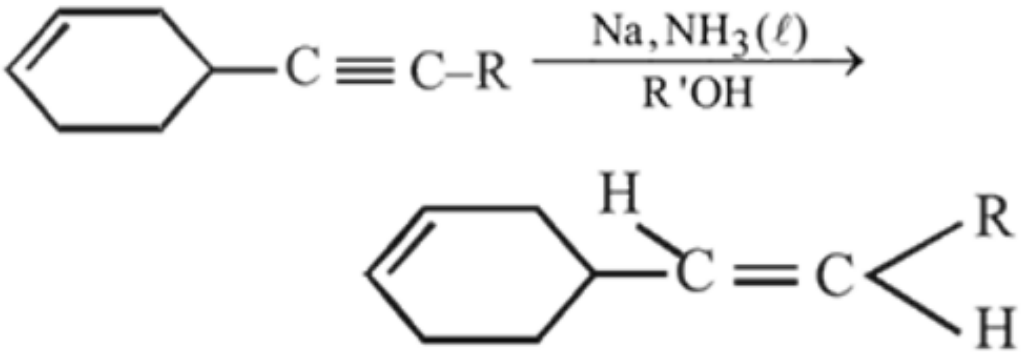
Q.18 Answer:



Solution:

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It is a streoselective reaction and in it a trans product is formed.



Q.19 Answer:

ammonical Cu_2Cl_2 solution

Solution:

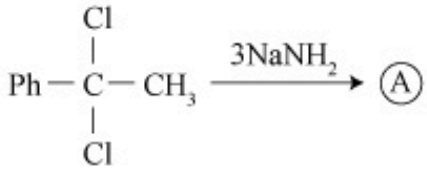
Ammonical cuprous chloride gives precipitate (red) only with terminal alkynes, i.e. alkynes having acidic hydrogen atom, such as butyne-1 $(CH \equiv CCH_2CH_3)$

Q.20 Answer:

$Ph - C \equiv \overset{\ominus}{C}\overset{\oplus}{N}$ a

Solution:

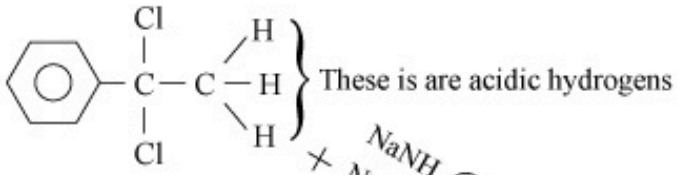
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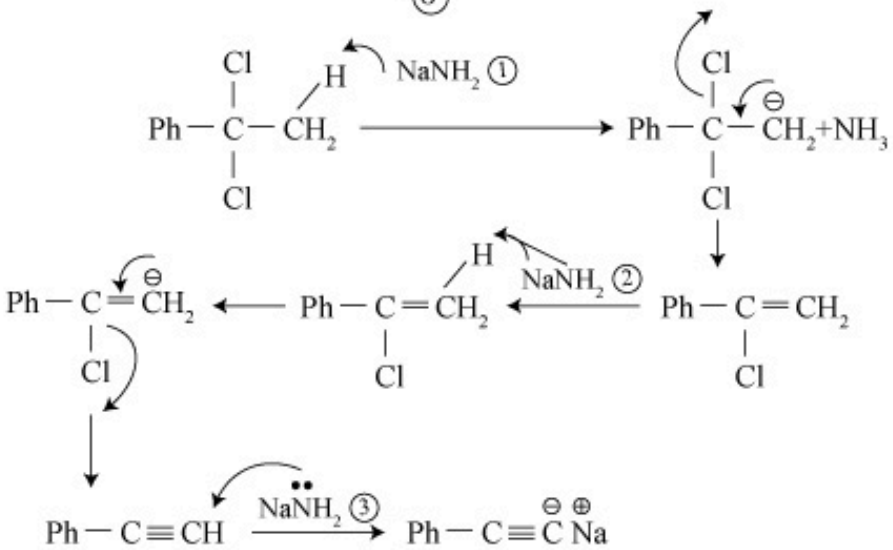


→ $NaNH_2$ is a strong box

→ It will deprotonate alkanes, alkenes, alkynes, etc.

→ $NaNH_2$ will depronates "acidic hydrogene"





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